National University of Computer and Emerging Sciences

Lab Manual 03

Programming Fundamentals

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Activities:

Activity 1:

Following basic program is shared for the understanding of students.

- Write a program that demonstrate the use of setw.

```
#include <iostream>
#include <iomanip>  //input output manipulation library
using namespace std;

int main()
{
    cout << setw(20) << "CodingisFun" << endl;
                          //setw is basically setting the width for text.
    cout << "Coding" << "is" << "fun" << endl;
    cout << setw(8) << "Coding" << setw(8) << "is" << setw(8) << "fun" << endl;

    return 0;
}
```

Activity 2:

Write the programs for the following:

1. Write a program that checks if a given integer is even or odd and prints the message accordingly.
2. Write a C++ program that takes two integer inputs (x1 and x2). Treat any non-zero value as true and zero as false. The program should then compute and print the XOR (exclusive OR) of these two values.

x1	x2	Output
False	False	False
False	True	True
True	False	True
True	True	False

Table 1: XOR Truth Table

Sample Output 1:
Enter the first number (x1): 5
Enter the second number (x2): 3
XOR of x1 and x2 is: 0

Sample Output 2:
Enter the first number (x1): 4
Enter the second number (x2): 0
XOR of x1 and x2 is: 1

3. Write a program that checks the temperature and prints if it is hot, warm, cool, or cold. Use the following temperature ranges:
   - Hot: 80°F or above
   - Warm: 60°F to 79°F
   - Cool: 40°F to 59°F
   - Cold: Below 40°F
4. Write a program that takes three integers as input and prints the largest of the three.
5. Write a C++ program that reads two integers from the user and compares them. Print "The first number is greater.", if the first number is greater than the second. Print "The second

number is greater.", if the second number is greater. Print "Both numbers are equal.", if they are the same.

6. Write a C++ program that reads a single character from the user and checks if it is a vowel (a, e, i, o, u). Print "The character is a vowel.", if it is a vowel; otherwise, print "The character is a consonant."
7. You are given 9 variables (x1, x2, x3, ..., x9) representing the slots of a 3x3 Tic-Tac-Toe game board. Each variable holds either 0 (representing player 1's move) or 1 (representing player 2's move). Write a C++ program that initializes these variables and then checks whether any player has won by completing a row or diagonal. If a player has won, print a message indicating which player has won. Handle only the rows and diagonals for simplicity.
   Here is the Tic-Tac-Toe grid:

x1	x2	x3
x4	x5	x6
x7	x8	x9

Table 2: Tic-Tac-Toe Grid

   Row Check: A player wins if all values in any row are the same (either 0 or 1). For Example, if x1, x2, and x3 have the same value, a player has won.
   Diagonal Check: A player wins if all values in a diagonal are the same (either 0 or 1). For Example, if x1, x5, and x9 have the same value, a player has won.
8. Write a single cout statement that prints the following 3 lines:

```
$$$
$"$"$
\$\$
```